

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid s$$

$$B \rightarrow b \mid \epsilon$$

Step 1: remove ϵ

nullable $\{A, B\}$

$$S \rightarrow ASA \mid SA \mid AS \mid s \mid aB \mid a$$

$$A \rightarrow B \mid s$$

$$B \rightarrow b$$

Step 2: remove unit productions

$$S \rightarrow ASA \mid SA \mid AS \mid s \mid aB \mid a$$

$$A \rightarrow ASA \mid SA \mid AS \mid s \mid aB \mid a \mid b$$

$$B \rightarrow b$$

After all the steps final Answer

$$S \rightarrow AX_1 \mid X_2SA \mid AS \mid X_3B \mid a$$

$$A \rightarrow AX_1 \mid SA \mid AS \mid X_3B \mid a \mid b$$

$$X_1 \rightarrow SA$$

$$B \rightarrow b$$

$$X_2 \rightarrow a$$

Que 16)

Convert the following grammar to CNF

$$S \rightarrow ABA$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$